

Controls

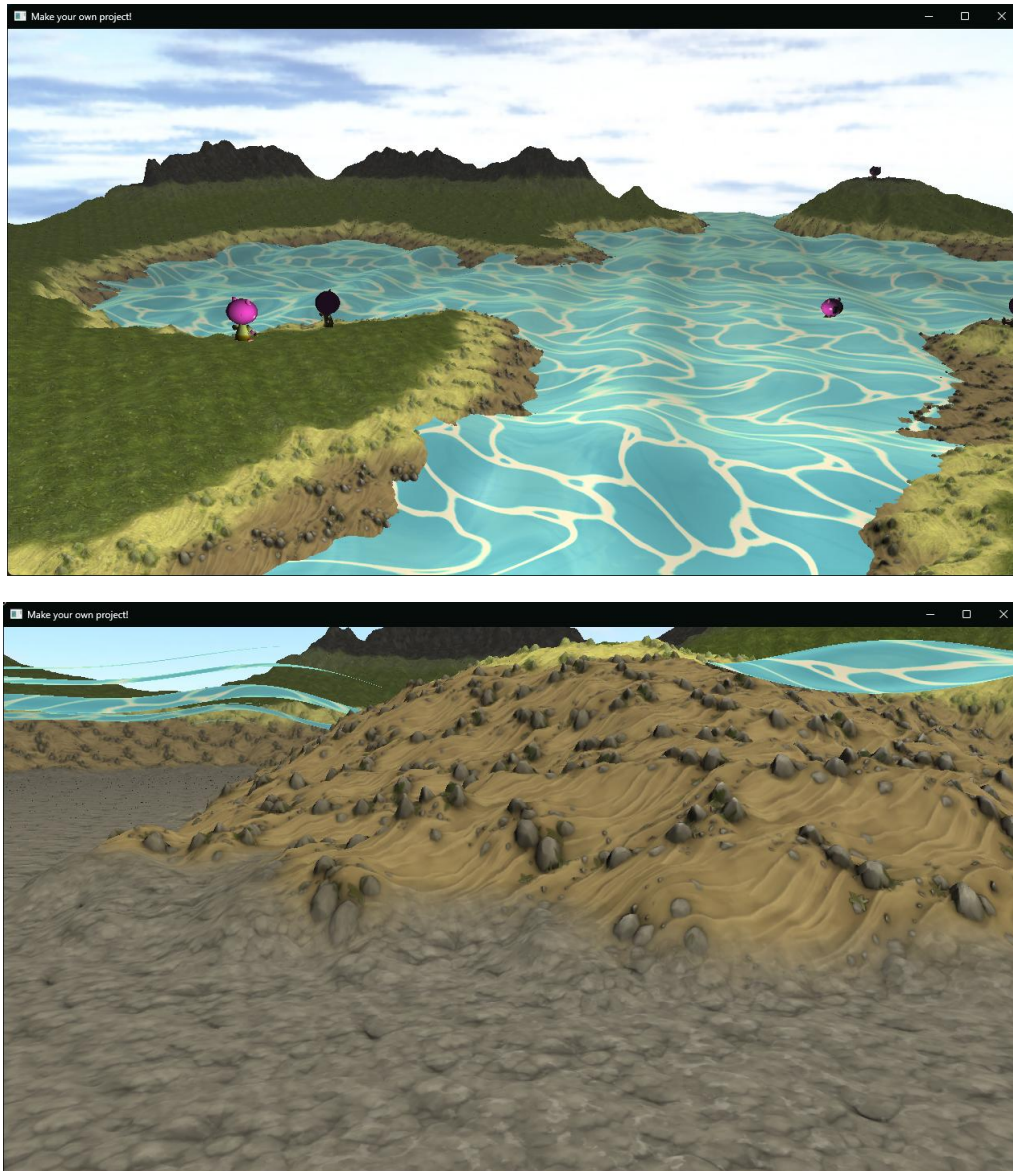
Enter – Swap between the controlled camera and free camera

Tab – Swap between scenes

Video Link:

<https://youtu.be/vArm3Q4p7QQ>

Scene Description



The first scene shows a sunny island environment with animated pink cats.

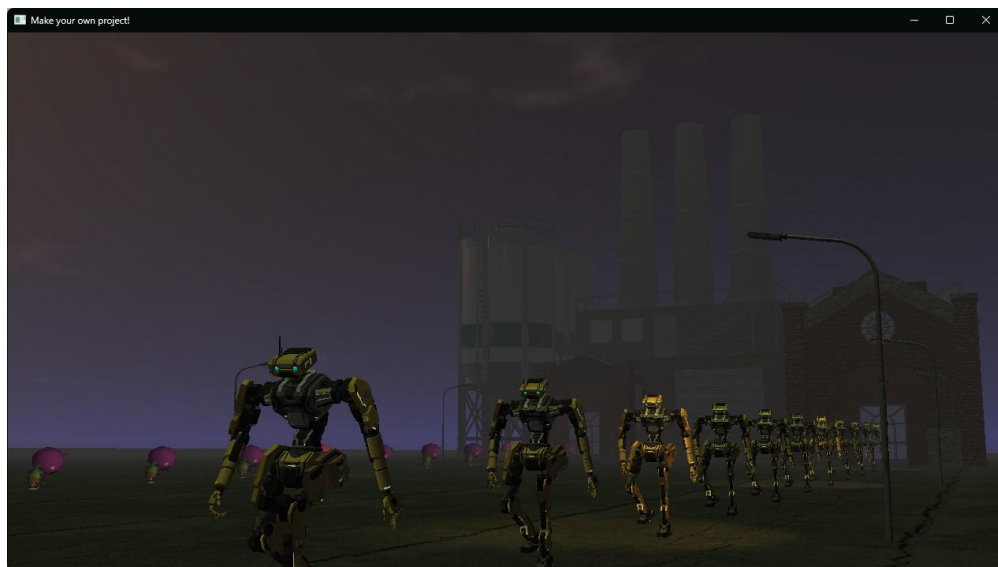
I've used a height map I drew and a tessellation shader to generate the terrain. I took four different PBR stylised materials and splat map I generated in GIMP to texture the ground. To do this I made a mesh material file to make loading in all the necessary files easier. The second image best showcases the blending between materials.

The materials I used had roughness, normal, ambient occlusion and height/displacement maps. To ensure the heights of the individual rocks in the sands and hill peaks were maintained, I passed in the displacement textures to the tessellation evaluation shader. Here, I sampled the textures using scaled down texture coordinates, passed through the vertex and control shaders. I multiplied the values sampled by a constant to control the contribution to the final height map.

The waves were also rendered using a tessellation shader. To achieve the effect of rolling waves, I applied compounded sine waves to the y values of each vertex, controlled by the x and z coordinates and time. At the same time, I worked out the gradients of the x and z waves, forming the tangent and bitangents, which I used to calculate the new normal for that vertex, and in turn calculate the reflections.

The pink cats in the scene were all loaded from copies of the global object loaded in the renderer. All of the animations used in this project are taken from Mixamo and all models are sourced from the internet. I made sure to find rigged models that worked well with Mixamo to simplify the process of matching up the bone names between the various files.

I downloaded each model as an fbx or gltf file and used the Assimp library to read in the data. I made various helper functions to ensure this data works with the nclgl library.



The second scene shows a line of the pink cats walking into a factory and a line of robots walking out. There are lamp posts that light up the environment.

I simulated the fog by increasingly adding the fog colour to fragments the further away they were from the camera.

The factory model I used had metallic textures. I applied these during the lighting pass, so shinier objects reflect more light. The reflection of the skybox can be seen on the silo in the image.

I chose to use spotlights to demonstrate the light cast by the lamp posts. I angled each of the casters slightly, so the light has softer edges at the points furthest away from the lamp post. This was achieved by implementing attenuation and simulating 2 cones of light being rendered, where the light inside the inner cone is more intense than the light outside it. Due to time constraints, I couldn't work out how to render the body of the cone to make the fog more obvious.

Furthermore, both scenes are also lit by directional lights, emulating the sun. I controlled the diffuse colour and specular highlights to create a brightly lit and warmer environment for the first scene, and a darker and grungier aesthetic for the second.

The robot model I used had an emissive texture. To add this, I added another colour attachment to the frame buffer so I could pass the colour sampled from the texture to the combine shader where I added it to the final fragment colour. To better show the effects of this, I multiplied it by a sine value controlled by time, so the lights on the robot pulse.

